

Notes on crop rotation optimization problem

1 Initial problem description and formulation

We consider the problem of finding an optimal allocation of crops to different fields for a single year in respect to maximizing the combined *profit*. Later on we will introduce additional constraints, that for example restricts the combined emission. The problem can without loss of generalization be swapped to minimizing emission with an constraint on economy, so one can think of profit as not necessarily in monetary terms, but some utility function we are interested in maximizing.

We denote the amount of fields by F and set of all fields $\mathbb{F} = \{1, 2, \dots, F\}$ and similarly the amount of different crops by C and set $\mathbb{C}$. The expected profit of planting a crop c on a field f is denoted $p_{c,f}$.

This can be formulated as the following *Integer Linear Program* (ILP):

$$\begin{aligned} \max_x \quad & \sum_{f=1}^F \sum_{c=1}^C p_{c,f} x_{c,f} \\ \text{s.t.} \quad & \sum_{c=1}^C x_{c,f} = 1 \quad \forall f \in \mathbb{F} \\ & x_{c,f} \in \{0, 1\} \quad \forall f \in \{1, \dots, F\}, c \in \mathbb{C} \end{aligned}$$

This is trivially solvable by selecting $x_{c',f} = 1$ for the c' that has $c_{c',f} = \max_{c \in \mathbb{C}} c_{c,f}$ for all $f \in \mathbb{F}$.

1.a Adding emission constraints

We introduce an emission value per field/crop choice denoted by $e_{c,f}$ and a hyperparameter $E \in \mathbb{N}$ to our optimization problem. E describes the maximum amount of emission we allow over across all fields. This extends the problem to:

$$\begin{aligned} \max_x \quad & \sum_{f=1}^F \sum_{c=1}^C p_{c,f} x_{c,f} \\ \text{s.t.} \quad & \sum_{c=1}^C x_{c,f} = 1 \quad \forall f \in \mathbb{F} \\ & \sum_{f=1}^F \sum_{c=1}^C e_{c,f} x_{c,f} \leq E \\ & x_{c,f} \in \{0, 1\} \quad \forall f \in \mathbb{F}, c \in \mathbb{C} \end{aligned}$$

Assuming this new constraint is violated in the optimal solution from before, this increases the complexity of the problem. However as the added constraint is linear we can still use modern ILP solvers to find near-optimal solutions in a reasonable time.

1.b Further constraints We have added a bunch of similar constraints such as

ensuring a minimum value of the total *nature value* of the fields and other linear constraints. They follow the same pattern as the emission constraint, and beside adding a bit of complexity to the solution space, they do not change the nature or type of optimization problem.

2 Extending the problem to multiple years

A big complexity in this project is planning multiple year ahead, because we want to model the fact that emission and yield for a given choice depends on the previous choice of crop.

We define $Y \in \mathbb{N}$ as the amount of years we want to plan and $\mathbb{Y}$ as the set of all years.

There are multiple ways this can be modelled.

2.a Linear modelling using transition decision variables

The first approach we tried was to introduce boolean decision variables for each field, year and crop transition possibility. Let $x_{y,\hat{c},c,f}$ denote the decision if we in year y plant crop c when the previous year we planted crop $\hat{c}$ on field f . We assume $x_{0,\hat{c},c,f}$ is known as a starting decision for all fields. Furthermore we define p and e with similar subscripts so $p_{\hat{c},c,f}$ is the profit of planting crop c on field f given that it last crop was $\hat{c}$

We can then formulate the problem as

$$\begin{aligned}
& \max_x \sum_{f=1}^F \sum_{y=1}^Y \sum_{c=1}^C \sum_{\hat{c}=1}^C p_{\hat{c},c,f} x_{y,\hat{c},c,f} \\
& \text{s.t.} : \sum_{c=1}^C \sum_{\hat{c}=1}^C x_{y,\hat{c},c,f} = 1 \quad \forall f \in \mathbb{F}, y \in \mathbb{Y} \\
& \sum_{f=1}^F \sum_{y=1}^Y \sum_{c=1}^C \sum_{\hat{c}=1}^C e_{\hat{c},c,f} x_{y,\hat{c},c,f} \leq E \\
& \sum_{c=1}^C x_{y,\hat{c},c,f} \leq \sum_{c=1}^C x_{y-1,\hat{c},\hat{c},f} \quad \forall f \in \mathbb{F}, y \in \mathbb{Y}, \hat{c} \in \mathbb{C} \\
& x_{y,\hat{c},c,f} \in \{0, 1\} \quad \forall f \in \mathbb{F}, c \in \mathbb{C}, \hat{c} \in \mathbb{C}, y \in \mathbb{Y}
\end{aligned}$$

This approach keeps the problem as an ILP, but the solution space is now $\mathbb{F} \times \mathbb{C} \times \mathbb{C} \times \mathbb{Y}$, and introduces $|\mathbb{F}| \cdot |\mathbb{Y}| \cdot |\mathbb{C}|$ new constraints, which for our use case makes the problem computationally unreasonable to solve.

2.b Quadratic optimization formulation

Alternatively we can formulate the decision variables as $x_{y,c,f}$ and modify the objective function as

$$\begin{aligned}
& \max_x \sum_{f=1}^F \sum_{y=1}^Y \sum_{c=1}^C \sum_{\hat{c}=1}^C p_{\hat{c},c,f} x_{y-1,\hat{c},f} x_{y,c,f} \\
& \text{s.t.} : \sum_{c=1}^C x_{y,c,f} = 1 \quad \forall f \in \mathbb{F}, y \in \mathbb{Y} \\
& \sum_{f=1}^F \sum_{y=1}^Y \sum_{c=1}^C \sum_{\hat{c}=1}^C e_{\hat{c},c,f} x_{y-1,\hat{c},f} x_{y,c,f} \leq E \\
& x_{y,\hat{c},c,f} \in \{0, 1\} \quad \forall f \in \mathbb{F}, c \in \mathbb{C}, \hat{c} \in \mathbb{C}, y \in \mathbb{Y}
\end{aligned}$$

This can also be rewritten to a quadratic form matrix optimization problem, however the problem now has quadratic constraints and objective, and is not guaranteed to be convex, so solving this is even more infeasible. We can note that the first problem archives its linear formulation by introducing the auxiliary variables equal to the product of $x_{y-1,\hat{c},f}$ and $x_{y,c,f}$

2.c Restricting the solution space

Going back to Section 2.a we can study the problem of the vast solution space. If we can define a reasonable set of valid crop rotations that can be chosen for each field we restrict the solution space a lot. Denote the set of all valid crop rotations over a period of Y years as $\mathbb{V}$. If we restrict that $|\mathbb{V}| \ll |\mathbb{C}| \cdot |\mathbb{C}| \cdot |\mathbb{Y}|$ we are back to the initial problem in Section 1 with $\mathbb{C}$ substituted with $\mathbb{V}$, which as long as $\mathbb{V}$ is of reasonable low cardinality we are able to find near-optimal solutions using modern solvers.